

QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3 – Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4 – Analyze]

Q3 Employee Salary Calculation [BL3 – Apply]

10 marks

Write pseudocode for an Employee class with encapsulated attributes (empld, name, basicPay, designation). Implement methods to:

Calculate gross salary (basic + HRA + DA)

- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding so salary components are not directly accessible.

OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
CLASS Employee
```

```
    PRIVATE empld
```

```
    PRIVATE name
```

```
    PRIVATE basicPay
```

```
    PRIVATE designation
```

```
    PRIVATE METHOD calculateHRA()
```

```
        RETURN basicPay * 0.20
```

```
    END METHOD
```

```
    PRIVATE METHOD calculateDA()
```

```
        RETURN basicPay * 0.15
```

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
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5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]

Your Answer

```
RETURN basicPay * 0.15
END METHOD

PRIVATE METHOD calculatePF()
    RETURN basicPay * 0.10
END METHOD

PRIVATE METHOD calculateTax()
    gross = calculateGrossSalary()
    RETURN gross * 0.05
END METHOD

PUBLIC METHOD init(empIdInput, nameInput, basicInput, designationInput)
    empId = empIdInput
    name = nameInput
    basicPay = basicInput
    designation = designationInput
END METHOD

PUBLIC METHOD calculateGrossSalary()
    hra = calculateHRA()
    da = calculateDA()
    RETURN basicPay + hra + da
END METHOD
```

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
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[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. University Staff Hierarchy
[BL4 – Analyze]

Your Answer

```
PUBLIC METHOD calculateNetSalary()
    gross = calculateGrossSalary()
    pf = calculatePF()
    tax = calculateTax()
    RETURN gross - (pf + tax)
END METHOD

PUBLIC METHOD displayPaySlip()
    hra = calculateHRA()
    da = calculateDA()
    gross = calculateGrossSalary()
    pf = calculatePF()
    tax = calculateTax()
    net = calculateNetSalary()

    PRINT "----- EMPLOYEE PAY SLIP -----"
    PRINT "Employee ID  :", empId
    PRINT "Employee Name :", name
    PRINT "Designation  :", designation
    PRINT "Basic Pay   :", basicPay
    PRINT "HRA (20%)   :", hra
    PRINT "DA (15%)    :", da
    PRINT "-----"
    PRINT "Gross Salary :", gross
    PRINT "PF (10%)    :", pf
```


Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

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– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

tax = calculateTax()

net = calculateNetSalary()

PRINT "----- EMPLOYEE PAY SLIP -----"

PRINT "Employee ID : ", empId

PRINT "Employee Name : ", name

PRINT "Designation : ", designation

PRINT "Basic Pay : ", basicPay

PRINT "HRA (20%) : ", hra

PRINT "DA (15%) : ", da

PRINT "-----"

PRINT "Gross Salary : ", gross

PRINT "PF (10%) : ", pf

PRINT "Tax (5% GST) : ", tax

PRINT "-----"

PRINT "Net Salary : ", net

PRINT "-----"

END METHOD

END CLASS

emp1 = NEW Employee()

emp1.init(192371057, "Naga Tejaswitha", 50000, "Software Engineer")

emp1.displayPaySlip()

 Save Answer